

Solve this 2D multi-span frame structure problem from scratch using the finite element method.

Problem Details:

- Geometry (all in mm): 3 bays with span $L = 6000$ mm per bay; 3 floors with floor height $H = 3000$ mm.
- Sections: Columns: rectangular 400×400 mm ($b \times h$, where b is out-of-plane width, h is in-plane height); Beams: rectangular 400×400 mm ($b \times h$, where b is out-of-plane width, h is in-plane height).
- Material: Linear elastic, $E = 3.0 \times 10^4$ MPa, $\nu = 0.3$; No body forces and consider shear deformation if necessary.

Nodal coordinates (x, y, unit: mm):

Total 16 nodes (4 positions in X, 4 in Y), with coordinates determined by spans/heights:

- X-direction: 3 spans $\times 6000$ mm $\rightarrow X = 0, 6000, 12000, 18000$ mm (4 positions: $i=1, 2, 3, 4$)
- Y-direction: 3 floors $\times 3000$ mm $\rightarrow Y = 0, 3000, 6000, 9000$ mm (4 positions: $j=1, 2, 3, 4$)

Element Division:

Total elements: 21 (12 columns + 9 beams).

Column elements (vertical, Y-direction): Connect nodes $(i, j) \rightarrow$ nodes $(i, j+1)$.

- Quantity: 4 (X positions: $i=1, 2, 3, 4$) $\times$ 3 (Y-intervals: $j=1 \rightarrow 2, 2 \rightarrow 3, 3 \rightarrow 4$) = 12.

Beam elements (horizontal, X-direction): Connect nodes $(i, j) \rightarrow$ nodes $(i+1, j)$.

- Quantity: 3 (X-intervals: $i=1 \rightarrow 2, 2 \rightarrow 3, 3 \rightarrow 4$) $\times$ 3 (floor positions: $j=2, 3, 4$) = 9.

Boundary Conditions:

- Bottom nodes: fully fixed ($u_x=0, u_y=0, \theta_z=0$);
- Rigid joints at all other nodes (shared u_x, u_y and θ_z); No other constraints.
- Load: Static horizontal concentrated force $F=50$ kN to the right (positive x-direction) at node (1, 2), node (1, 3) and node (1, 4); No other loads.

Calculations required:

- Nodal Displacement and rotation: u_x, u_y , and θ_z ;
- Axial force N (kN), shear force V (kN), and bending moment M (kN·mm) diagrams for each element;
- Provide visualization of corresponding deformation diagrams and internal force diagrams.

Programming language: Python 3.10.

Permitted libraries: NumPy 2.1.1, SciPy 1.14.1, JAX 0.4.30, Matplotlib 3.9.2.